Dynamic Signal Effects: The Role of Social Media in Box Office Prediction

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Abstract

With the increasing importance of social media in the film industry for box office prediction and promotional strategy formulation, this study employs signaling theory and social influence theory to explore its role in box office performance. Leveraging web crawling technology by Python, this study collected data from multiple social media platforms, extracting film-related signals such as information volume, platform ratings, and comprehensive heat index. Using SPSS software for multiple linear regression analysis, an empirical study was conducted on 256 Chinese films released between 2018-2022 with box office revenues exceeding one million yuan. The results demonstrate that incorporating social media indicators significantly enhances the explanatory power of the prediction model. Specifically, the volume of information, platform ratings, and comprehensive heat index positively influence box office revenue. This research integrates computer-aided data collection and statistical analysis methods, providing both theoretical contributions to signaling theory and practical insights for the film industry's digital transformation.

CCS Concepts

• **Applied computing** → Law, social and behavioral sciences; Economics.

Keywords

Box office, social media, box office prediction, signaling theory

ACM Reference Format:
Jinhan Shi, Haodong Gu, and Aixin Chen. 2025. Dynamic Signal Effects: The Role of Social Media in Box Office Prediction. In *2025 2nd International Conference on Digital Society and Artificial Intelligence (DSAI 2025), May 23–25, 2025, Wuhan, China.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3748825.3748971

1 Introduction

In the digital era, social media has emerged as a pivotal force in shaping consumer behavior and influencing market outcomes, particularly in the film industry where box office performance is critical to commercial success. As an experiential product, films face significant information asymmetry before release, which is the situation in which one party has less information than the other in a transaction as audiences rely on pre-release signals to gauge quality and decide whether to watch [1, 2]. Traditional signals, such as director reputation or star power, have long been studied, but the rise of social media has introduced a new paradigm of dynamic, user-generated signals that can rapidly spread and amplify consumer sentiment [3]. Identifying these market signals is crucial for box office research. For instance, the Chinese New Year film *NE ZHA 2* surpassed 15.7 billion yuan, ranking 5th on the list of Worldwide Box Office Ranking at the end of April 2025. The growth of the film industry is closely tied to social media signals. From a marketing perspective, *NE ZHA 2* exemplifies how modern promotional strategies—such as leveraging social media or shaping electronic word-of-mouth (eWOM) to influence audience perceptions and viewing decisions—have become indispensable in contemporary film marketing.

Currently, the film industry has transitioned from traditional marketing models to the digital marketing era. Social media, as a new digital channel, has become one of the key indicators influencing audience movie decisions and box office predictions [4]. However, despite the growing recognition of social media's role, existing research has primarily focused on text mining or sentiment analysis, often neglecting the dynamic interplay between different social media signals and their cumulative impact on box office revenue [5, 6]. While studies have shown that user reviews and trailer views correlate with box office performance [7], the mechanisms through which these signals interact and evolve over time remain under explored. Moreover, research on predicting box office performance by tracking movie-related information on social media remains insufficient, which can significantly influence audience perception and drive last-minute decisions, which are pivotal for opening-week performance and long-term success [8].

In order to better promote the sustainable development of the film economy, an important approach is to accurately predict box office performance. Thus, the purpose of this study is to investigate the dynamic signal effects of social media in box office prediction, focusing on how three key dimensions—volume of information, platform ratings, and comprehensive heat index—influence box office revenues. By integrating insights from signaling theory and social influence theory, we aim to uncover how these signals collectively mitigate information asymmetry and drive audience behavior. The following sections will review relevant literature on movie box office predicting, its technical approaches, and the moderating roles of affective congruence and topic consistency. The research hypotheses will then be proposed, followed by an explanation of the

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DSAI 2025, Wuhan, China
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1433-7/2025/05
https://doi.org/10.1145/3748825.3748971

materials and method, the results, and a discussion of the study's implications.

2 Literature Review

2.1 Signaling Theory

Signaling theory is particularly useful for analysing consumers, who are often in an information asymmetry position in the market [9]. In such situations, consumers are unable to directly assess the quality of a product, so they rely on the signals they receive from the market to gauge product quality. Information asymmetry leads consumers to lack concrete perceptions when facing products and services they have not experienced, thereby affecting their willingness to purchase. In the context of film signalling theory, information asymmetry is a fundamental issue for choosing whether to watch the movie or not. Stiglitz [10] emphasized two types of information asymmetry: quality information and intent information. The former involves one party's ignorance about the attributes and features of the other party, while the latter pertains to the lack of understanding regarding the behaviour and intentions of the other party. Firstly, research found that consumers use other indicators within products and services, such as brand and performance, as well as status signals, to assess quality [11]. Advertising and recommended prices can serve as indirect quality signals for emerging consumer goods [12]. Furthermore, when consumers lack sufficient information, they use brand signals to infer the quality attributes of products, thereby positioning and evaluating them. The credibility of brand signals is positively correlated with consumers' perceptions of product quality; the more credible the signal, the better the consumer's perception of quality.

In addition to traditional signals such as price and brand advertising, there are a variety of signals in the online environment that convey product quality, with word-of-mouth (WOM) being one of the most important. Liu [13] proposed that electronic word-of-mouth (eWOM) is a collective reputation signal, becoming the preferred source for consumers seeking purchase experience information. Consumer purchase decisions are influenced by two types of social signals: one being the purchasing behaviour of peer consumers, and the other being their reviews [14]. Among these signals, peer reviews are usually defined as eWOM, which plays a key role in the decision-making process of consumers [15].

2.2 Social Influence

Social influence is defined as "the changes in an individual's thoughts, emotions, attitudes or behaviors resulting from interaction with others or groups" [16, 17]. Social influence theory (SIT) posits that individuals' behavior is shaped by the real or imagined influence of others [18]. In the context of movie box office prediction, this theory provides a crucial framework for understanding how social media signals impact audience decisions. Research has shown that social influence significantly affects consumer choices, and in the movie - watching context, it plays a pivotal role in determining box office success [19].

Social influence describes the phenomenon of individual belief, attitude, emotion, or behaviour change under the influence of social forces [20]. These social forces can originate from individuals or groups. First, social influence can lead to conformity behavior, where an individual changes their original stance or behaviour under the influence of others to align with the group's behaviour. Deutsch [18] noted that individuals use information from others as real evidence, thus enhancing their understanding of the environment and their ability to respond to specific situations. Second, informational social influence can occur in two ways: 1) individuals acquire information and knowledge from others; 2) they make judgments by observing others' behaviours [21]. This type of influence is particularly evident in consumers' product evaluation and purchasing decisions. Kim [22] argued that this is essentially a learning process, where consumers in the online environment observe the experiences of other buyers to decide whether to purchase a product.

In the context of digital marketing, SIT has been used to explain user behavior on social media platforms [23]; and the attitudinal and behavioural implications of online influencer marketing [24]. With the widespread use of the internet, eWOM and similar informational influences play a crucial role in shaping and guiding consumer behaviour. Numerous empirical studies indicate that persuasive information like eWOM can have a significant impact on individuals' behaviour, even triggering behavioural changes [25, 26]. Compared to the past, when social circles were smaller and social influences were more limited, the internet has vastly expanded the range of social influence through social media, allowing an individual's influence to spread more widely. Social influence not only helps build a product's reputation but also deeply impacts the recommendation process, thereby affecting consumers' trust [27].

2.3 Social Media and Movie Box Office Prediction

Social media has increasingly played a critical role in the movie production process, especially in box office forecasting. Movie box office performance is a crucial indicator of a film's commercial success, and accurately predicting it can help stakeholders in the film industry make informed decisions. Accurate movie box office forecasting is critical for industry stakeholders, yet existing research faces challenges in prediction accuracy and data complexity. Early studies using neural networks incorporated factors like star power and ratings but achieved limited accuracy due to simplistic models and narrow data scope [28]. Later work, such as Barman [29] highlighted the role of complex interfering factors—like piracy [30], which reduces theater demand—but noted scalability issues in real-world applications. These limitations stem from flawed data acquisition, over-simplified models, and inadequate consideration of interactive variables.

Many studies focus on extracting various signals from social media to predict movie box office performance. Forecasting methods broadly fall into two categories: statistical models and machine learning. Methods based on statistical models, like regression and time series models, face challenges such as assuming linear relationships or relying too much on historical data. Although machine learning and deep-learning are applied [31, 32], the empirical research is still insufficient [32, 33]. Statistical approaches, such as regression and time-series analysis, rely on historical data but are constrained by linearity assumptions and narrow scope

[34, 35]. Others evaluate film efficiency by Data Envelopment Analysis (DEA) [36]. Machine learning methods, including neural networks and support vector machines, excel at capturing complex relationships but often underutilize comprehensive data, limiting precision despite promising potential. Some research applied data or text mining tools, and big data, combining multi-source social media and web data like IMDb movie databases, to extract decision factors like film genres, director and follower counts [5, 6, 37], others combined movie sentiment analysis [37-39]. Researchers have predominantly used data from single social media platforms such as Twitter [39], Facebook, or Microblogs [2, 37, 40]. Existing research on movie box office forecasting has limitations in data acquisition, prediction accuracy, and limited platforms [40, 41]. In summary, the analysis and mining of social media data have become an essential method for forecasting movie box office performance. This not only helps film production companies develop more effective market strategies but also provides audiences with a more personalized viewing experience.

Existing studies have examined the role of social media in predicting movie box office performance from various perspectives, covering prediction methods. However, there are still some issues that require further research. For instance, most studies are based on data from specific regions or time periods, and the universality of their results needs verification; when considering social media signals, the study of synergistic effects among different platforms is insufficient; and further exploration is needed on how to better integrate multiple influencing factors to improve the accuracy of box office predictions. In box office forecasting research, the selection and analysis of movie signals are crucial. Based on the factors mentioned in existing literature, we can summarize box office prediction research into three main aspects. First, directly using box office revenue as a forecasting indicator, such as predicting future box office performance based on a movie's first-day or first-week performance. Second, focusing on internal movie signals, such as the cast, director, plot, story content, and ticket prices, to predict the overall box office performance. Finally, considering external signals, especially social media information such as reviews, likes, and sentiment in tweets, which provide additional perspectives for box office forecasting. Introducing external signals from social media in the research process can enhance the explanatory power of forecasting models and analytical processes. Therefore, this paper will focus on the application of social media indicators, as external signals, in box office forecasting based on signal theory.

3 Hypotheses

Social media has become one of the most important channels for promoting and marketing movies. The volume of information on social media, which comes from official data sets on social media platforms, effectively expands a movie's audience, increases its exposure, and reflects feedback from social media on the movie. Movie-related information released on official social media platforms, such as likes, shares, and views, can rapidly spread across cyberspace [42]. This provides valuable reference information for assessing the movie's performance in the market. Before making a purchase decision, consumers often search for relevant product information. During this process, consumers can receive movie signals from official sources, which profoundly influence their movie perception. Combined with social influence theory, information from highly credible sources has a more significant impact on consumer behaviour and attitudes [43]. When consumers receive substantial official feedback from trusted social media platforms, it is likely to influence their ticket-buying decisions. Abundant social media-related information can attract more attention to the movie, guide the public's views, and increase box-office performance. The related videos received a large number of likes and shares on platforms such as Weibo and WeChat. After encountering this information, many social media users were eager to purchase tickets, which led to a dramatic increase in the film's box office performance, demonstrating the powerful impact of official social media information on movie box office success. If a movie has abundant social media-related information, the movie enjoys widespread audience awareness and intense topic discussion on social media platforms. This is likely to attract more attention to the movie. Therefore, the information disseminated via social media has the potential to guide the public in forming specific views about the movie and sparking interest. When a large volume of positive information about a movie emerges on social media, it is likely to motivate more users to develop an interest in watching the film, thereby driving an increase in box office performance and enhancing the movie's commercial competitiveness and influence. Therefore, we hypothesize the following:

Hypothesis 1 (H1): The volume of information about a movie on social media positively influences its box office revenue.

In today's world, movie ratings on social media platforms have become an extremely important evaluation criterion in modern culture. With the ongoing evolution and maturation of social media, people are increasingly accustomed to exploring, interacting, and sharing various types of information on these platforms. Platform ratings as a key indicator that is likely to have a significant impact on movie box office performance. Platform ratings are a specific form of eWOM and are highly persuasive [26, 44]. Compared to analysing the positive and negative emotional effects in online reviews, movie ratings are easier to collect and come with clear standards for measurement, which makes them a commonly used indicator for assessing movie word-of-mouth in various studies. Platform ratings come from the movie's audience, i.e., the vast number of movie consumers or platform users, and they accurately reflect these audiences' direct impressions of the movie, which can influence the ticket-buying decisions of potential viewers. The relationship between online ratings, movie revenues, and viewer satisfaction are verified as a U sharp [45]. While some scholars have studied movie ratings on social media, most are based on data from a single social media platform or single type of rating (Rated R) [46], leading to considerable discrepancies in their conclusions. By selecting rating data from multiple social media platforms and integrating them into a platform rating, the influence mechanism between ratings and box office performance. Platform ratings are derived from actual viewing experiences and reviews from the audience, and whether they show a positive or negative trend will influence the choices and decisions of other potential viewers, thus affecting the movie choice and revenue. Therefore, we hypothesize the following:

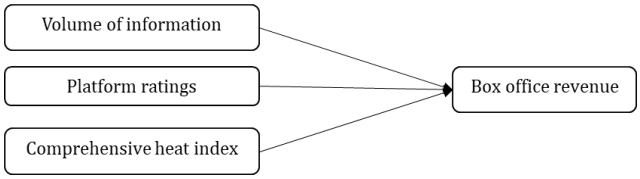


Figure 1: The conceptual model of this study.

Hypothesis 2 (H2): The platform ratings of a movie on social media positively influence its box office revenue.

With the rapid development of online social media, people are increasingly inclined to use various channels to obtain the latest movie information, related reviews, and actively participate in interactions. The data recorded from these interactions often deeply reflect the audience's preferences and expectations for the movie. Social media's proliferation has amplified film-related information consumption on platforms like IMDb and YouTube [47, 48]. Drawing from signal theory, active information-seeking behaviors expose consumers to relevant cues, thereby facilitating information dissemination and reducing purchase uncertainty [49]. The comprehensive heat index combines various indicators such as trailer heat, "want-to-see" heat, media heat, Weibo heat, WeChat heat, and search index through specific calculation methods. The integrated heat index comprehensively reflects the overall movie signals across different social media platforms and objectively evaluates the attitudes and confidence levels of these platforms by summarizing multi-channel heat to accurately show the movie's attention and discussion levels. The heat displayed by the movie on social media platforms highlights its influence within a specific range and also exerts a certain degree of impact on box office performance. Some indicators in the integrated heat index reflect the strong intention of users to actively search for and express their interest in watching the movie. The hotter the interest, the higher the corresponding heat, which also indicates a stronger demand for the movie. Additionally, the integrated heat index provides an intuitive reflection of the audience's strong interest and eager anticipation for the film. Therefore, the comprehensive heat index not only reflects the movie's exposure level and user attention but also embodies people's demand preferences and feedback, thus exerting a significant influence on the film's box office performance. Therefore, we hypothesize the following:

Hypothesis 3 (H3): The comprehensive heat index of a movie on social media positively influences its box office revenue.

4 Materials and Methods

4.1 Research Framework

Based on the hypotheses below, Figure 1 shows the conceptual model of our study.

4.2 Sample Selection and Data Collection

This study collected domestic box office data for films released in China from 2018 to 2022, sourced from Box Office Mojo, and selected 256 films that achieved box office earnings of over one million RMB. These films were used to construct the data sample for in-depth analysis. As a globally recognized box office statistics website, Box Office Mojo provided key data support for this research. The five-year period (2018-2022) was strategically chosen to capture the dynamic characteristics of the current film market, which ensures that the research findings are more aligned with contemporary market trends. This period aligns with the rapid pace of digital transformation in China's film industry. Social media platforms (such as Douban and Weibo) have experienced exponential user growth, while streaming services (such as Bilibili and Tencent Video) have begun to exert significant influence on theatrical release strategies. Meanwhile, the selected time frame includes a critical market inflection point—the COVID-19 pandemic (2020), which reshaped audience behavior and distribution models. The pandemic accelerated the shift of films toward online promotion. Therefore, this period is well-suited for exploring the impact of social media on film box office performance while reflecting future trends, demonstrating strong practical significance. The selection of films with box office earnings above one million RMB was made because such films tend to attract larger audiences and exert a broader market influence. Analysing these films provides more representative and reliable data samples, forming a solid foundation for exploring trends in film-related indicators and identifying key factors affecting box office performance.

In this study, web crawling technology was primarily employed for data collection by Python, which involves using automated programs to simulate human browsing behavior and extract publicly available data from the internet. It enables the efficient acquisition of information dispersed across diverse platforms, making it particularly suitable for research requiring large-scale, structured data support, such as social media analysis and industry trend insights.

The relevant variables in this study were mainly derived from publicly available data from major official websites, including Box Office Mojo, Weibo, WeChat, Bilibili, Douyin, Douban, Taopiaopiao, Maoyan, Lighthouse Professional Edition, and Baidu. Specifically, data were collected on Weibo's likes, shares, comments, and topic discussions, as well as the number of followers for official accounts, actors, and promotional song artists. WeChat data included the reading volumes of official movie accounts. Bilibili data focused on view counts, while Douyin provided data on topic playback volumes. Rating data were obtained from Douban and Taopiaopiao, while Maoyan provided detailed box office data, ticket prices, production companies, and distribution companies, along with information on movie length, showtimes, and historical box office performance of actors, directors, and screenwriters. The Lighthouse Professional Edition provided the Popcorn Index, and Baidu data included basic information about actors, directors, and screenwriters such as their gender, age, filmography, and awards, as well as Baidu Index data. For data with large values, such as box office earnings, production and distribution company earnings, opening week showtimes, and the number of followers for promotional song artists, logarithmic transformations were applied to reduce large disparities in data, thereby improving the accuracy and scientific rigor of the analysis. The volume of information, platform ratings, and data related to actors, directors, and screenwriters were subjected to factor analysis for dimensionality reduction, yielding more representative and comprehensive variables.

4.3 Variable Measurement

In this study, the dependent variable was defined as the box office revenue, which refers to the total accumulated box office revenue during their release period. This indicator directly reflects the overall commercial performance of the films. Typically, the higher the box office earnings, the more viewers chose to watch the film, implying greater audience recognition and favourability, and demonstrating its commercial value. For the purpose of this study, the box office earnings were subjected to logarithmic transformation to enhance the precision and scientific rigor of subsequent analysis.

4.3.1 Volume of Information. Weibo, WeChat, Bilibili, and Douyin all have large user bases and broad audience reach. As major social media platforms, they provide highly credible and authoritative content. Therefore, this study performed factor analysis for dimensionality reduction on the information volume of films in their first week of release across these four platforms and presented the results as a composite "volume of information" indicator. Specifically, Weibo's information volume was derived through factor analysis of the official account's followers, likes, shares, topic readings, and discussions, which capture various forms of film-related Weibo content. Bilibili's information volume was based on the top three films by view count, while WeChat's information volume was derived from the top three films by official account reading volume, and Douyin's information volume was directly obtained from topic playback volume. These four types of data mostly stem from official guidance and promotion, accurately reflecting the film's popularity and the sustainability of its spread across social media platforms. In essence, the independent variable of volume of information effectively captures the feedback signals from social media, providing a crucial basis for exploring the relationship between films and social media.

4.3.2 Platform Ratings. This study selected the first-week ratings from Maoyan, Taopiaopiao, and Douban as the key indicators for the independent variable of platform ratings. Maoyan and Taopiaopiao are the two largest online ticket platforms in China, while Douban is a highly influential fan community and film database in the country. Other rating platforms were excluded due to issues such as foreign user ratings, incomplete data due to copyright restrictions, and small user bases. The ratings from Maoyan, Taopiaopiao, and Douban collectively provide a comprehensive view of social media platform ratings. All three platforms use a ten-point rating system, combining user ratings with professional reviews. These ratings accurately reflect the film's quality and popularity from the social media user perspective. In this study, the first-week ratings were selected, as they are more likely to influence potential viewers' movie selection and ticket purchasing decisions. Ratings obtained after the film has been off-screen may not be as impactful for driving box office performance. Therefore, the first-week ratings were chosen to explore their intrinsic relationship with box office earnings.

4.3.3 Comprehensive Heat Index. In this study, the Popcorn Index is used to measure the comprehensive heat index. The Popcorn Index is a more comprehensive indicator that integrates data from authoritative platforms through detailed analysis. Specifically, the Lighthouse Professional Edition calculates this index by assigning weights to various internal data sources, including news media, Weibo, WeChat, "want-to-see" heat, trailer heat, search heat, etc. The resulting index offers a thorough presentation of the film's exposure and audience attention across multiple platforms, providing a key quantifiable basis for studying a film's market performance and influence. The Popcorn Index helps analyse the relationship between the film's popularity and other key factors, including box office earnings.

4.3.4 Control Variables. To improve the accuracy and reliability of the results and clearly illustrate the influence and mechanism of social media on box office performance, control variables were introduced. These variables stem from key signals associated with the film itself. The study selected ten control variables: price, production company, distribution company, film length, prequels, opening-week showtimes, music, lead actors, directors, and screenwriters. Specifically, the average ticket price refers to the average price during the film's release period. For production and distribution companies, the logarithm of the total box office earnings of films produced and distributed by these companies before the current film was used. Film length was determined based on the movie's runtime. The prequel variable is coded as 1 if the film is a sequel or has a prequel, and 0 if not. The opening-week showtimes were transformed using logarithms. The music variable was calculated as the logarithm of the promotional song artist's followers on Weibo. For the lead actors, directors, and screenwriters, various relevant metrics were selected, and factor analysis was used for dimensionality reduction to create a composite measure.

4.3.5 Variable Summary. The names, detailed descriptions, and data sources of variables in this study are summarized in Table 1.

Due to the substantial differences in magnitude between the variables of movie box office revenue, producer, distributor, opening-week showtimes, and movie theme song compared to other variables, it is necessary to preprocess these data prior to conducting correlation tests and multiple linear regression analyses. This study employs logarithmic transformation to modify the original values to meet the requirements of the analysis. By applying natural logarithm transformation, the range of the variables is appropriately reduced without affecting the recognizability and interpretability of their fundamental relationships. Additionally, because the variables of volume of information, platform ratings, lead actor, director, and screenwriter contain numerous similar feature indicators, factor analysis is required for these types of data. Factor analysis is utilized to eliminate redundant features, reduce errors, improve data quality, and simplify the data analysis process. Furthermore, our research analysis indicates that social media data exhibit significant correlations. Therefore, extracting principal components and major variances can help capture core information for box office forecasting.

The following variables were subjected to factor analysis using SPSS software:

Volume of information: Factor analysis was conducted using the official Weibo followers, likes, shares, topic readings, and topic discussions in the first week of the film's release (KMO value = 0.752, $\chi 2 = 952.756$, $p < 0.001$). The top three Bilibili view counts were reduced to the Bilibili information volume factor (KMO value = 0.787, $\chi 2 = 3004.019$, $p < 0.001$). The top three WeChat official

Table 1: Variable Description.

Variable category	Variable name	Description	Data Source
Dependent variable	Movie box office revenue	Total box office earnings during the film's release period, with logarithmic transformation	Box Office Mojo
Independent variable	Volume of information	Fans, likes, shares, comments, topic readings, and topic discussions on official Weibo; view count on Bilibili; reading volume on WeChat official accounts; topic playback volume on Douyin, with principal component analysis	Weibo, WeChat, Bilibili, Douyin
	Platform ratings	Film ratings in the first week of release on Douban, Taopiaopiao, and Maoyan, with principal component analysis	Douban, Taopiaopiao, Maoyan
	Comprehensive heat index	Popcorn Index	Lighthouse Professional Edition
Control variable	Average movie price	Average ticket price during the film's release period	Maoyan
	Producer	Cumulative total box office earnings of films produced by the producer before the current film, with logarithmic transformation	Maoyan
	Distributor	Cumulative total box office earnings of films distributed by the distributor before the current film, with logarithmic transformation	Maoyan
	Duration	Film runtime	Maoyan
	Sequel	Whether the film is a sequel (Binary Variable: 0 = No, 1 = Yes)	Douban
	Opening-week showtimes	Number of showtimes in the first week of release, with logarithmic transformation	Maoyan
	Movie theme song	Number of Weibo fans of the movie's theme song artist, with logarithmic transformation	Weibo
	Lead actor	Historical box office of the lead actor, Weibo followers, Baidu Index on the release day, awards, and number of films participated in, with principal component analysis	Maoyan, Weibo, Baidu
	Director	Baidu Index of the director on the release day, historical box office of the director, number of films directed, awards, with principal component analysis	Maoyan, Baidu
	Screenwriter	Historical box office of the screenwriter, number of works participated in, awards, with principal component analysis	Maoyan, Baidu

account reading volumes were reduced to the WeChat information volume factor (KMO value = 0.729, $\chi 2 = 3139.949$, $p < 0.001$). Finally, the topic playback volumes from Weibo, Bilibili, WeChat, and Douyin were further subjected to factor analysis, resulting in the independent variable "Volume of Information" (KMO value = 0.640, $\chi 2 = 198.457$, $p < 0.001$).

Platform ratings: Factor analysis was performed on movie ratings from Douban, Taopiaopiao, and Maoyan during the first week of release (KMO value = 0.653, $\chi 2 = 737.104$, and $p < 0.001$).

Lead actor: Factor analysis was conducted on the lead actor's historical box office, Weibo followers, Baidu Index on the day of film release, awards, and number of films participated in (KMO value = 0.676, $\chi 2 = 1246.772$, $p < 0.001$).

Director: Factor analysis was conducted on the director's Baidu Index on the day of film release, historical box office, number of films directed, and awards (KMO value = 0.704, $\chi 2 = 467.443$, $p < 0.001$).

Screenwriter: Factor analysis was performed on the screenwriter's historical box office, number of works participated in, and awards (KMO value = 0.645, $\chi 2 = 192.283$, $p < 0.001$).

5 Results

In this study, a systematic analysis of variables was conducted by comprehensively applying computational techniques based on the SPSS software platform. Descriptive statistical analysis was used to quantitatively characterize data features, correlation analysis was employed to uncover the degree of association between variables, and linear regression analysis was utilized to establish causal relationship models among variables, thereby deeply exploring the internal connections and interaction mechanisms between variables.

Firstly, this study performed descriptive statistical analysis on the dependent variable and independent variables. The result shows that movie box office revenue (Mean = 19.187, SD = 1.481); volume of information (Mean =1.244, SD = 1.103); platform ratings (Mean

Table 2: The correlation result of variables.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Average movie price	1													
Producer	0.137*	1												
Distributor	-0.028	0.181**	1											
Duration	0.233**	0.042	0.072	1										
Sequel	0.05	0.102	-0.024	-0.013	1									
Opening-week showtimes	0.130*	0.302**	0.112	0.273**	0.064	1								
Movie theme song	0.156*	0.171**	0.092	-0.006	-0.029	0.308**	1							
Lead actor	0.195**	0.112	-0.027	0.251**	-0.135*	0.352**	0.174**	1						
Director	0.133*	0.145*	0.024	0.221**	-0.036	0.182**	0.016	0.314**	1					
Screenwriter	0.093	0.082	-0.005	0.149*	-0.032	0.046	-0.178**	0.203**	0.531**	1				
Volume of information	0.379**	0.180**	0.071	0.298**	0.155*	0.463**	0.241**	0.118	0.089	-0.009	1			
Platform ratings	0.160*	0.032	0.090	0.241**	-0.068	0.230**	-0.056	0.013	0.055	0.078	0.235**	1		
Comprehensive heat index	0.080	0.193**	0.033	-0.057	0.023	0.400**	0.237**	0.168**	0.098	0.019	0.132*	0.037	1	
Movie box office revenue	0.409**	0.334**	0.100	0.228**	0.154*	0.816**	0.356**	0.333**	0.149*	-0.013	0.561**	0.315**	0.454**	1

Note: *p < 0.05, **p < 0.01, ***p < 0.001

= 4.416, SD =1.014); comprehensive heat index (Mean = 70.114, SD = 23.935). Notably, the standard deviation of the comprehensive heat index is relatively large, indicating considerable variation in social media activity across different films. A further examination shows that the gap between the maximum and minimum values is quite significant. This phenomenon highlights the stark differences in the level of social media engagement between films. Some films generate considerable buzz on social media platforms, attracting substantial user attention and discussion, while others are relatively ignored, creating a significant disparity in popularity.

Secondly, correlation analysis was conducted on the dependent variable, independent variables, and control variables. The correlation coefficient matrix can be found in Table 2. As can be seen from the correlation coefficients, the volume of information, platform ratings and comprehensive heat index all have a significantly positive correlation with movie box office revenue (r = 0.561, p < 0.01; r = 0.315, p < 0.01; r = 0.454, p < 0.01), which preliminarily validates hypotheses H1, H2, and H3.

Thirdly, this study conducted linear regression analysis to further validate the hypotheses. In this study, two regression models were constructed as shown in the formulas of Model 1 and Model 2 below.

In Model 1, only control variables and dependent variables were included.

$$Movie\ Box\ Office\ Revenue_i = \beta_0 + \sum_{s=1}^{s} \beta_1^s Movie\ Control_{s,i} + \delta_i + \varepsilon_i \qquad (Model1)$$

In Model 2, independent variables were added on the basis of Model 1.

$$Movie\ Box\ Office\ Revenue_i = \beta_0 + \sum_{s=1}^{s} \beta_1^s Movie\ Control_{s,i} + \beta_2 Volume\ of\ Information_i + \beta_3 Platform\ Ratings_i + \beta_4 Comprehensive\ Heat\ Index_i + \delta_i + \varepsilon_i \qquad (Model2)$$

To ensure the scientific rigor and accuracy of the multiple regression model, variance inflation factor (VIF) diagnostics were conducted for all the variables included in the model (VIF < 5), indicating that there is no severe multicollinearity among the independent variables and control variables. This result effectively eliminates any potential interference or negative impact on the prediction model caused by multicollinearity, providing a solid and reliable data foundation for subsequent analyses and conclusion derivations based on this model.

The result of Model 1 shows that the average movie ticket price has a positive effect on box office (Table 3). This is primarily because ticket prices are endogenous, often determined by factors such as production costs, movie quality, and market popularity. Additionally, given a fixed number of viewers, higher average ticket prices naturally lead to higher total box office revenue, resulting in a strong correlation between the two. Compared to non-sequel movies, sequels tend to attract higher popularity and box office revenue. Many studies have provided explanations for this phenomenon. Future research could further explore the relationship between adaptations and films.

Table 3: Result of Regression Estimation.

Variable	Model 1	Model 2
(Constant)	-2.708**	0.639
Average movie price	0.100***	0.081***
Producer	0.013	0.012
Distributor	0.006	0.003
Duration	-0.004	-0.005*
Sequel	0.268**	0.276**
Opening-week showtimes	1.428***	1.179***
Movie theme song	0.025	0.022
Lead actor	0.052	0.089*
Director	0.006	0.005
Screenwriter	-0.082*	-0.086**
Volume of information		0.149***
Platform ratings		0.189***
Comprehensive heat index		0.008***
R^2	0.782	0.820
Adjusted R^2	0.774	0.810
F Value	88.128***	84.555***

Note: *p < 0.05, **p < 0.01, ***p < 0.001

We focus on discussing the influence of social media on movie box office revenue. A comparison of Model 1 and Model 2 shows that Model 1 is significant ($F = 88.128$, $p < 0.001$), with an R^2 of 0.782 and an adjusted R^2 of 0.774, indicating a relatively high explanatory power. After including social media-related variables, Model 2 remains significant ($F = 84.555$, $p < 0.001$), with an R^2 of 0.820 and an adjusted R^2 of 0.810, both of which are higher than those in Model 1, with a statistically significant change.

Upon examining the regression coefficients of the three independent variables in Model 2, the result shows that social media volume of information, platform ratings and comprehensive heat index all have a significant positive impact on movie box office revenue ($\beta = 0.149$, $p < 0.001$; $\beta = 0.189$, $P < 0.001$; $\beta = 0.008$, $P < 0.001$). Thus, the results validate hypotheses H1, H2 and H3.

6 Discussion

In this study, we conducted an in-depth analysis of box office data from 256 movies, testing the hypotheses regarding the mechanisms through which three key social media signals—volume of information, social media platform ratings, and comprehensive heat index—affect movie box office performance. Through a series of scientific analyses and validations, we found that all three hypotheses hold true. This will provide stronger theoretical support and practical reference for research and decision-making in the film industry.

6.1 Theoretical Contributions

On one hand, the findings of this study offer several significant theoretical contributions by integrating and extending signal theory and social influence theory in the context of movie box office performance. Building upon signaling theory, which posits that observable indicators can reduce information asymmetry, we demonstrate that social media metrics act as dynamic market signals that reliably convey information about a film's appeal and quality to potential audiences. In parallel, drawing from social influence theory, our results confirm that aggregated online behaviors (such as ratings and engagement levels) serve as persuasive social proof that shapes individual consumer decisions [14]. By merging these two perspectives, the research introduces a unified framework in which social media indicators are conceptualized as credible signals emanating from both official channels and the audience community, thereby redefining the role of social media in film industry research. This theoretical integration expands the current understanding of movie success drivers by highlighting how digital word-of-mouth and promotional signals jointly contribute to box office performance. On the other hand, this study develops and validates a comprehensive predictive model that incorporates multi-dimensional social media metrics alongside traditional predictors. Previous research on box office forecasting often examined limited factors or single-platform data, leading to fragmented insights [39,46]. In contrast, our model simultaneously accounts for official information volume, audience-driven ratings, and comprehensive heat index that aggregates cross-platform engagement, capturing a more holistic picture of a movie's market signal profile. This finding extends signaling theory by empirically validating that novel, non-traditional signals (e.g., the cross-platform heat index) carry significant informational value in predicting real-world outcomes.

6.2 Practical Implications

In addition to theoretical advancements, the study yields practical implications that are highly relevant for film industry stakeholders, including producers, marketing professionals, and social media platform operators. On one hand, for film producers and marketing professionals, the study highlights that investments in social media strategies are not merely ancillary promotions but critical

components of a film's commercial success. Ensuring a high volume of timely, official information releases across popular social networks helps build early awareness and acts as a positive signal to moviegoers. Likewise, fostering strong eWOM can amplify social influence effects that persuade undecided audiences to purchase tickets [49]. Marketing campaigns should integrate social media analytics as a core tool: by monitoring metrics such as the information volume and heat index in real time, practitioners can gauge audience sentiment and engagement levels, enabling them to adjust advertising spend or promotional tactics on the fly. On the other hand, for social media platform operators, the demonstrated link between platform data (e.g., post views, shares, likes, and user ratings) and box office performance highlights the value of maintaining an authentic and engaging user environment. Platforms that curb fake accounts, bot-driven interactions, or malicious rating behavior will provide more reliable signals to the market, which in turn encourages studios to engage more deeply with those channels for publicity. There is also incentive for platforms to collaborate with content producers to elevate a movie's visibility (its "heat") in the weeks around release.

6.3 Limitations and Future Research

This study utilized a relatively small sample size of 256 films, which may impact the accuracy and reliability of the findings. A larger sample size is needed to enhance the completeness and precision of the data, ensuring that the research conclusions are robust. Additionally, the objectivity of platform ratings is not guaranteed, as users may engage in biased behaviours, such as malicious or repeated ratings. This subjective error could affect the data. Future research could incorporate diverse perspectives and additional social media indicators, such as sentiment analysis, consumer behavior, and brand value, to enhance prediction accuracy. Lastly, this study focused on the correlation between social media and first-tier distribution channels but did not consider the entire distribution process, including secondary (and illegal) channels. Future research could explore the role of social media in multi-tiered movie distribution, capturing its broader influence throughout other moderators.

7 Conclusions

This study, leveraging web crawling technology by Python and SPSS-based multiple linear regression analysis, systematically explores the impact of social media on box office prediction within the framework of signaling and social influence theories. Empirical analysis of 256 Chinese films (2018-2022) reveals that social media information volume, platform ratings, and comprehensive heat index significantly enhance the predictive accuracy of box office revenue. These results highlight the dual role of digital platforms as both communication channels and behavioral amplifiers in the film industry. By integrating machine-assisted data collection and statistical learning methods, this research bridges traditional economic models with cutting-edge computational techniques, providing actionable insights for optimizing promotional strategies and resource allocation. As social media continues to reshape audience engagement patterns, its integration with AI-driven analytics will remain pivotal for fostering data-driven decision-making and ensuring the sustainable development of the film ecosystem.

Acknowledgments

This research was funded by the General Project of Shanghai Philosophy and Social Science, Grant no: 2020BGL031.

References

[1] Mavlanova, T.; Benbunan-Fich, R.; Koufaris, M. Signaling theory and information asymmetry in online commerce. *Information & Management* **2012**, *49*, 240-247. https://doi.org/10.1016/j.im.2012.05.004.
[2] Du, J.; Xu, H.; Huang, X. Box office prediction based on microblog. *Expert Systems with Applications* **2014**, *41*, 1680-1689. https://doi.org/10.1016/j.eswa.2013.08.065.
[3] Fan, L.; Zhang, X.; Rai, L. When should star power and eWOM be responsible for the box office performance? - An empirical study based on signaling theory. *Journal of Retailing and Consumer Services* **2021**, *62*, 102591. https://doi.org/10.1016/j.jretconser.2021.102591.
[4] Hennig-Thurau, T.; Ravid, S.A.; Sorenson, O. The Economics of Filmed Entertainment in the Digital Era. *Journal of Cultural Economics* **2021**, *45*, 157-170. https://doi.org/10.1007/s10824-021-09407-6.
[5] Hwangbo, H.; Kim, J. A Text Mining Approach for Sustainable Performance in the Film Industry. *Sustainability* **2019**, *11*, 3207. https://doi.org/10.3390/su11113207.
[6] Apala, K.R.; Jose, M.; Motnam, S.; Chan, C.-C.; Liszka, K.J.; Gregorio, F.d. Prediction of movies box office performance using social media. In Proceedings of the 2013 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining, Niagara, Ontario, Canada, **2013**; pp. 1209–1214. https://doi.org/10.1145/2492517.25002
[7] Lee, Y.-J.; Keeling, K.B.; Urbaczewski, A. The Economic Value of Online User Reviews with Ad Spending on Movie Box-Office Sales. *Information Systems Frontiers* **2019**, *21*, 829-844. https://doi.org/10.1007/s10796-017-9778-7.
[8] Rubin, D.; Mohr, I.; Kumar, V. Beyond the box office: A conceptual framework for the drivers of audience engagement. *Journal of Business Research* **2022**, *151*, 473-488. https://doi.org/10.1016/j.jbusres.2022.07.021.
[9] Xu, X.-Y.; Jia, Q.-D. A new exploration of signaling theory in social commerce facilitated cross-border retailing: A four-stage approach. *Information & Management* **2025**, *62*, 104154. https://doi.org/10.1016/j.im.2025.104154.
[10] Stiglitz, J.E. The Contributions of the Economics of Information to Twentieth Century Economics. *The Quarterly Journal of Economics* **2000**, *115*, 1441-1478. https://doi.org/10.1162/003355300555015.
[11] Pandey, S.; Mittal, S.; Chawla, D. Tackling consumer information asymmetry and perceived uncertainty for luxury re-commerce through seller signals. *Journal of Retailing and Consumer Services* **2024**, *79*, 103736. https://doi.org/10.1016/j.jretconser.2024.103736.
[12] Fluet, C.; Garella, P.G. Advertising and prices as signals of quality in a regime of price rivalry. *International Journal of Industrial Organization* **2002**, *20*, 907-930. https://doi.org/10.1016/S0167-7187(01)00075-3.
[13] Liu, H.; Jayawardhena, C.; Shukla, P.; Osburg, V.-S.; Yoganathan, V. Electronic word of mouth 2.0 (eWOM 2.0) – The evolution of eWOM research in the new age. *Journal of Business Research* **2024**, *176*, 114587. https://doi.org/10.1016/j.jbusres.2024.114587.
[14] Cheung, C.M.K.; Xiao, B.S.; Liu, I.L.B. Do actions speak louder than voices? The signaling role of social information cues in influencing consumer purchase decisions. *Decision Support Systems* **2014**, *65*, 50-58. https://doi.org/10.1016/j.dss.2014.05.002.
[15] Rani, A.; Shivaprasad, H.N. Revisiting the antecedent of electronic word-of-mouth (eWOM) during COVID-19 Pandemic. *DECISION* **2021**, *48*, 419-432. https://doi.org/10.1007/s40622-021-00298-2.
[16] Liang, Q.; Liao, X.; Liu, J. A social ties-based approach for group decision-making problems with incomplete additive preference relations. *Knowledge-Based Systems* **2017**, *119*, 68-86. https://doi.org/10.1016/j.knosys.2016.12.001.
[17] Monteserin, A.; Amandi, A. Whom should I persuade during a negotiation? An approach based on social influence maximization. *Decision Support Systems* **2015**, *77*, 1-20. https://doi.org/10.1016/j.dss.2015.05.003.
[18] Deutsch, M.; Gerard, H.B. A study of normative and informational social influences upon individual judgment. *The Journal of Abnormal and Social Psychology* **1955**, *51*, 629-636. https://doi.org/10.1037/h0046408.
[19] Feng, N.; Shi, Y.; Li, Y.; Li, D.; Zhang, J.; Li, M. An Exploration of the Dynamics Between Social Media and Box Office Performance. *Information Systems Frontiers* **2024**, *26*, 591-608. https://doi.org/10.1007/s10796-023-10389-3.
[20] Smith, J.; Louis, W.; Schultz, P. Introduction: Social influence in action. *Group Processes & Intergroup Relations - GROUP PROCESS INTERGROUP RELA* **2011**, *14*, 599-603. https://doi.org/10.1177/1368430211410214.
[21] Park, C.W.; Lessig, V.P. Students and Housewives: Differences in Susceptibility to Reference Group Influence. *Journal of Consumer Research* **1977**, *4*, 102-110. https://doi.org/10.1086/208685.
[22] Kim, Y.A.; Srivastava, J. Impact of social influence in e-commerce decision making. In Proceedings of the ACM International Conference Proceeding Series, **2007**; pp. 293-302. https://doi.org/10.1145/1282100.1282157.

[23] Oliveira, C.; Garcia, A.C.B.; Vivacqua, A.S. The cost structure of influencers' posts: the risk of losing followers. *Personal and Ubiquitous Computing* **2021**, *25*, 259-280. https://doi.org/10.1007/s00779-020-01502-3.

[24] Tafesse, W.; Wood, B.P. Followers' engagement with instagram influencers: The role of influencers' content and engagement strategy. *Journal of Retailing and Consumer Services* **2021**, *58*, 102303. https://doi.org/10.1016/j.jretconser.2020.102303.

[25] Ismagilova, E.; Dwivedi, Y.K.; Slade, E.; Williams, M.D. Persuasiveness of eWOM Communications. In *Electronic Word of Mouth (eWOM) in the Marketing Context: A State of the Art Analysis and Future Directions*, Ismagilova, E., Dwivedi, Y.K., Slade, E., Williams, M.D., Eds.; Springer International Publishing: Cham, **2017**; pp. 49-71. Available online: https://link.springer.com/chapter/10.1007/978-3-319-52459-7_5.

[26] Tsao, W.-C.; Hsieh, M.-T. eWOM persuasiveness: do eWOM platforms and product type matter? *Electronic Commerce Research* **2015**, *15*, 509-541. https://doi.org/10.1007/s10660-015-9198-z.

[27] Zhao, L.; Xu, Y.; Xu, X. The effects of trust and platform innovation characteristics on consumer behaviors in social commerce: A social influence perspective. *Electronic Commerce Research and Applications* **2023**, *60*, 101284. https://doi.org/10.1016/j.elerap.2023.101284.

[28] Sharda, R.; Delen, D. Predicting box-office success of motion pictures with neural networks. *Expert Systems with Applications* **2006**, *30*, 243-254. https://doi.org/10.1016/j.eswa.2005.07.018.

[29] Barman, D.; Singha, R.K.; Chowdhury, N. Prediction of possible business of a newly launched film using ordinal values of film-genres. *International Journal of Intelligent Systems and Applications (IJISA)* **2013**, *5*, 53-60. https://doi.org/10.5815/ijisa.2013.06.07.

[30] Chisholm, D.C.; Fernández-Blanco, V.; Abraham Ravid, S.; David Walls, W. Economics of motion pictures: The state of the art. *Journal of Cultural Economics* **2015**, *39*, 1-13. https://doi.org/10.1007/s10824-014-9234-1.

[31] Li, D.; Liu, Z.-P. Predicting Box-Office Markets with Machine Learning Methods. *Entropy* **2022**, *24*, 711. https://doi.org/10.3390/e24050711.

[32] Menaga, D.; Lakshminarayanan, A. A Method for Predicting Movie Box-Office using Machine Learning. In Proceedings of the 2023 4th International Conference on Electronics and Sustainable Communication Systems (ICESC), 6-8 July 2023, **2023**; pp. 1228-1232. https://doi.org/10.1109/ICESC57686.2023.10192928.

[33] Withdraw, W. withdraw. *Applied and Computational Engineering* **2024**, *100*, 156-161. https://doi.org/10.54254/2755-2721/100/20251729.

[34] Chintagunta, P.K.; Gopinath, S.; Venkataraman, S. The effects of online user reviews on movie box office performance: Accounting for sequential rollout and aggregation across local markets. *Marketing science* **2010**, *29*, 944-957. https://doi.org/10.1287/mksc.1100.0572.

[35] Lee, Y.; Kim, S.-H.; Cha, K.C. A generalized Bass model for predicting the sales patterns of motion pictures having seasonality and herd behavior. *Journal of Global Scholars of Marketing Science* **2012**, *22*, 310-326. https://doi.org/10.1080/21639159.2012.721214.

[36] Lee, S.; Choeh, J. Movie Production Efficiency Moderating between Online Word-of-Mouth and Subsequent Box Office Revenue. Sustainability, **2020**, 12 (16), 6602. https://doi.org/10.3390/su12166602.

[37] Ahmad, I.S.; Bakar, A.A.; Yaakub, M.R. Movie Revenue Prediction Based on Purchase Intention Mining Using YouTube Trailer Reviews. *Information Processing & Management* **2020**, *57*, 102278. https://doi.org/10.1016/j.ipm.2020.102278.

[38] Sadadi, H.; Aloufi, D.; Ye, Z. Predict Movie Revenue by Sentimental Analysis of Twitter. In Proceedings of the International Conference on Data Processing and Applications, Guangdong, China, **2018**; pp. 1–4. https://doi.org/10.1145/3224207.3224208.

[39] Kwon, B. Twitter Sentiment Analysis and Box Office Revenue Prediction. **2021**. https://doi.org/10.17615/cyk5-xh18.

[40] Qiu, X.; Tang, T.Y. Microblog Mood Predicts the Box Office Performance. In Proceedings of the 2018 Artificial Intelligence and Cloud Computing Conference, **2018**; pp. 129-133. https://doi.org/10.1145/3299819.3299839.

[41] Song, T.; Huang, J.; Tan, Y.; Yu, Y. Using User- and Marketer-Generated Content for Box Office Revenue Prediction: Differences Between Microblogging and Third-Party Platforms. *Information Systems Research* **2019**. https://doi.org/10.1287/isre.2018.0797.

[42] Gupta, S.; Deodhar, S.J.; Tiwari, A.A.; Gupta, M.; Mariani, M. How consumers evaluate movies on online platforms? Investigating the role of consumer engagement and external engagement. *Journal of Business Research* **2024**, *176*, 114613. https://doi.org/10.1016/j.jbusres.2024.114613.

[43] Hu, X.; Chen, X.; Davison, R.M. Social support, source credibility, social influence, and impulsive purchase behavior in social commerce. *International Journal of Electronic Commerce* **2019**, *23*, 297-327. https://doi.org/10.1016/j.jbusres.2024.114613.

[44] Chen, H.; Duan, W.; Zhou, W. When products receive reviews across platforms: Studying the platform concentration of electronic word-of-mouth. *Information & Management* **2021**, *58*, 103532. https://doi.org/10.1016/j.im.2021.103532.

[45] Moon, S.; Bergey, P.K.; Iacobucci, D. Dynamic Effects among Movie Ratings, Movie Revenues, and Viewer Satisfaction. *Journal of Marketing* **2010**, *74*, 108-121. https://doi.org/10.1509/jmkg.74.1.108.

[46] Palsson, C.; Price, J.; Shores, J. Ratings and revenues: Evidence from movie ratings. *Contemporary Economic Policy* **2013**, *31*, 13-21. https://doi.org/10.1111/j.1465-7287.2012.00315.x.

[47] Schwemmer, C.; Ziewiecki, S. Social media sellout: The increasing role of product promotion on YouTube. *Social Media+ Society* **2018**, *4*, 2056305118786720. https://doi.org/10.1177/2056305118786720.

[48] Chiu, Y.-L.; Chen, K.-H.; Wang, J.-N.; Hsu, Y.-T. The impact of online movie word-of-mouth on consumer choice: a comparison of American and Chinese consumers. *International Marketing Review* **2019**, *36*, 996-1025. https://doi.org/10.1108/IMR-06-2018-0190.

[49] Arif, M.; Siregar, I.K. Purchase decision affects, price, product quality and word of mouth. *International Journal of Economic, Technology and Social Sciences (Injects)* **2021**, *2*, 260-273. https://doi.org/10.53695/injects.v2i1.389.